

DSC 40B

Lecture 11 : Sorting.

MergeSort

MergeSort

Can we sort faster?

- The *tight theoretical lower bound* for **comparison** sorting is $\Theta(n \log n)$.
- Selection sort is **quadratic**.
- How do we sort in $\Theta(n \log n)$ time?

Mergesort

- Mergesort is a **fast** sorting algorithm.
- Has **best possible** (worst-case) time complexity: $\Theta(n \log n)$.
- Implements **divide/conquer/recombine** strategy.

The Idea

- **Divide**: split the array into halves
 - $[6,1,9,2,4,3] \rightarrow [6,1,9], [2,4,3]$
- **Conquer**: sort each half, recursively
 - $[6,1,9] \rightarrow [1,6,9]$ and $[2,4,3] \rightarrow [2,3,4]$
- **Combine**: merge sorted halves together
 - $[1,6,9], [2,3,4] \rightarrow [1,2,3,4,6,9]$

Aside: splitting arrays

- Splitting an array in half by **slicing**:

```
>>> arr = [9, 1, 4, 2, 5]
```

```
>>> middle = math.floor(len(arr) / 2)
```

```
>>> arr[:middle]
```

```
[9, 1]
```

```
>>> arr[middle:]
```

```
[4, 2, 5]
```

- **Warning! Creates a copy!**

Python 3.6
([known limitations](#))

```
1 arr = [9, 1, 4, 2, 5]
2 middle = len(arr) // 2
3 first_half = arr[:middle]
→ 4 second_half = arr[middle:]
```

[Edit this code](#)

executed
execute

<< First

< Prev

Next >

Last >>

Done running (4 steps)

[ation](#)

Frames

Global frame

arr

middle

first_half

second_half

2

Objects

list

0	1	2	3	4
9	1	4	2	5

list

0	1
9	1

list

0	1	2
4	2	5

Aside: splitting arrays

- Splitting an array in half by **slicing**:

```
>>> arr = [9, 1, 4, 2, 5]
```

```
>>> middle = math.floor(len(arr) / 2)
```

```
>>> arr[:middle]
```

```
[9, 1]
```

```
>>> arr[middle:]
```

```
[4, 2, 5]
```

- **Warning! Creates a copy!**

Time complexity for slicing?

A: Constant

B: Log

C: Linear

D: Quadratic

Aside: splitting arrays

- Splitting an array in half by **slicing**:

```
>>> arr = [9, 1, 4, 2, 5]
```

```
>>> middle = math.floor(len(arr) / 2)
```

```
>>> arr[:middle]
```

```
[9, 1]
```

```
>>> arr[middle:]
```

```
[4, 2, 5]
```

- **Warning! Creates a copy!**

Time complexity for slicing?

A: Constant

B: Log

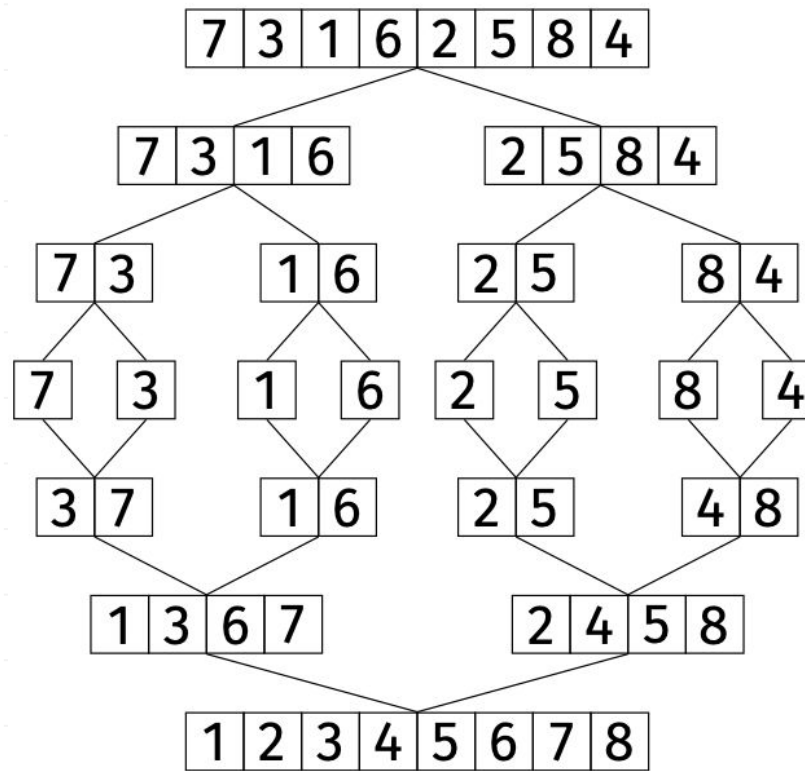
C: Linear

D: Quadratic

Mergesort

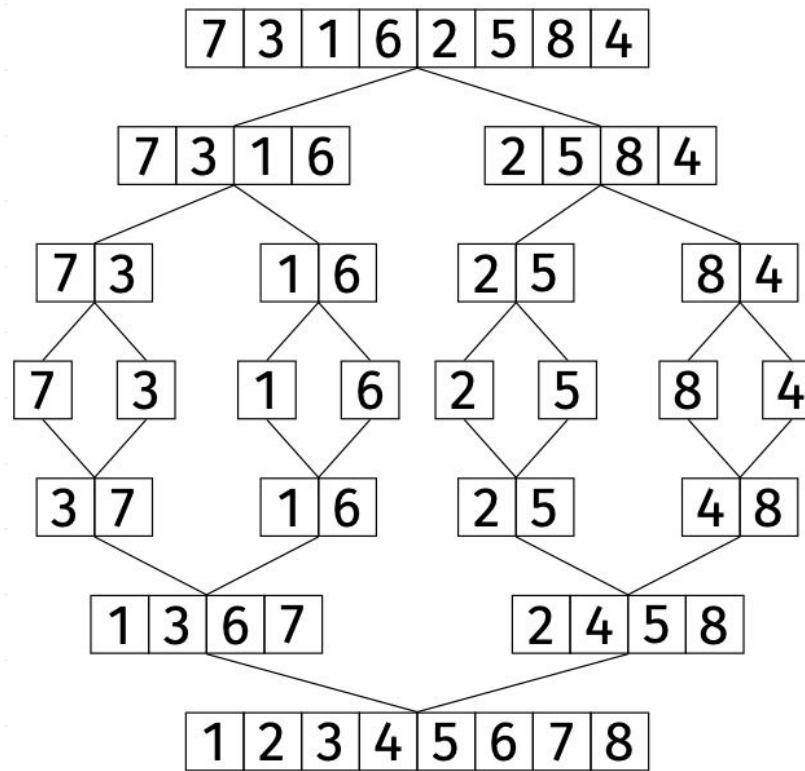
```
def mergesort(arr):  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        mergesort(left)  
        mergesort(right)  
        merge(left, right, arr)
```

The Idea



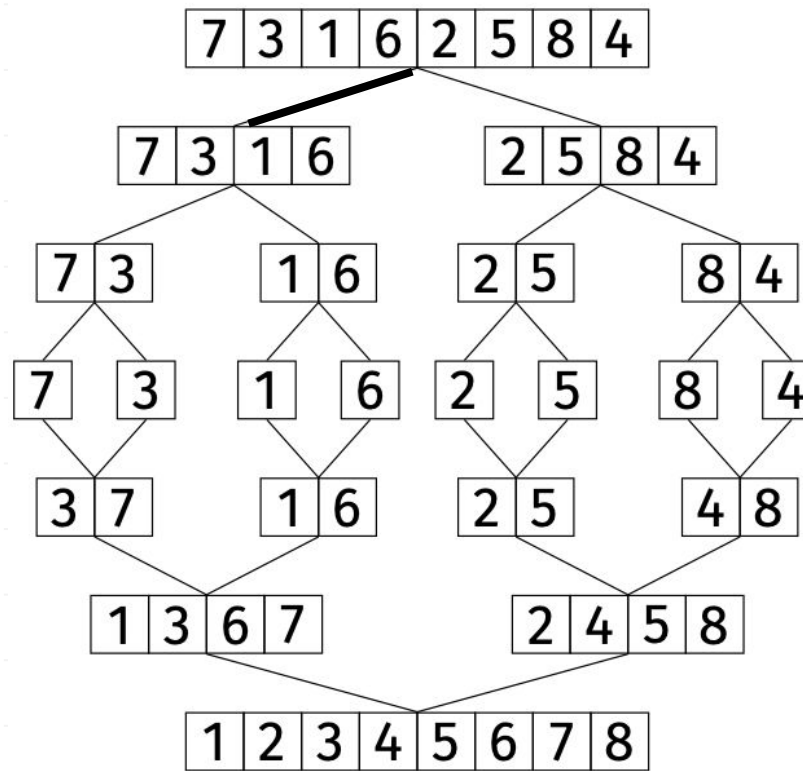
`ms([7, 3, 1, 6, 2, 5, 8, 4])`

The Idea



ms([7, 3, 1, 6, 2, 5, 8, 4])

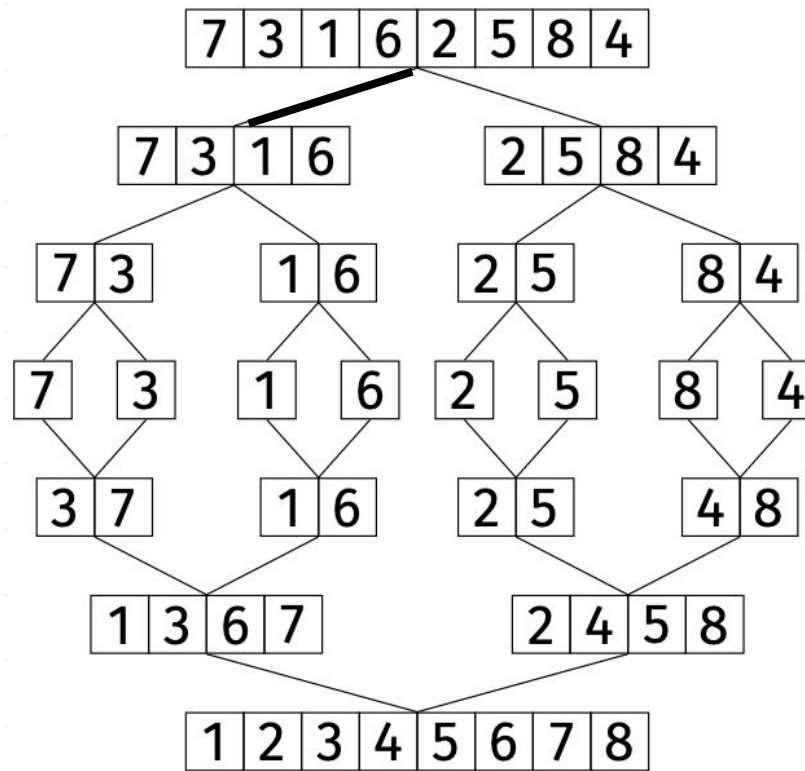
The Idea



$\text{ms}([7, 3, 1, 6, 2, 5, 8, 4])$

$\text{ms}([7, 3, 1, 6])$

The Idea



`ms([7, 3, 1, 6, 2, 5, 8, 4])`

`ms([7, 3, 1, 6])`

What is happens next?

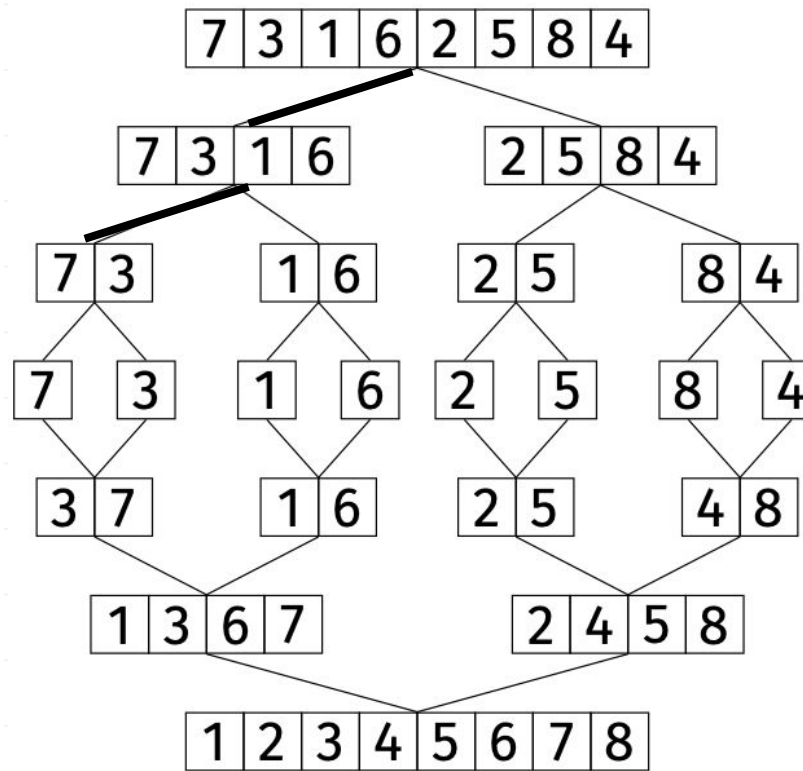
A: `ms([7, 3])`

B: `ms([2, 5, 8, 4])`

C: `ms([1, 6])`

D: starts to merge

The Idea

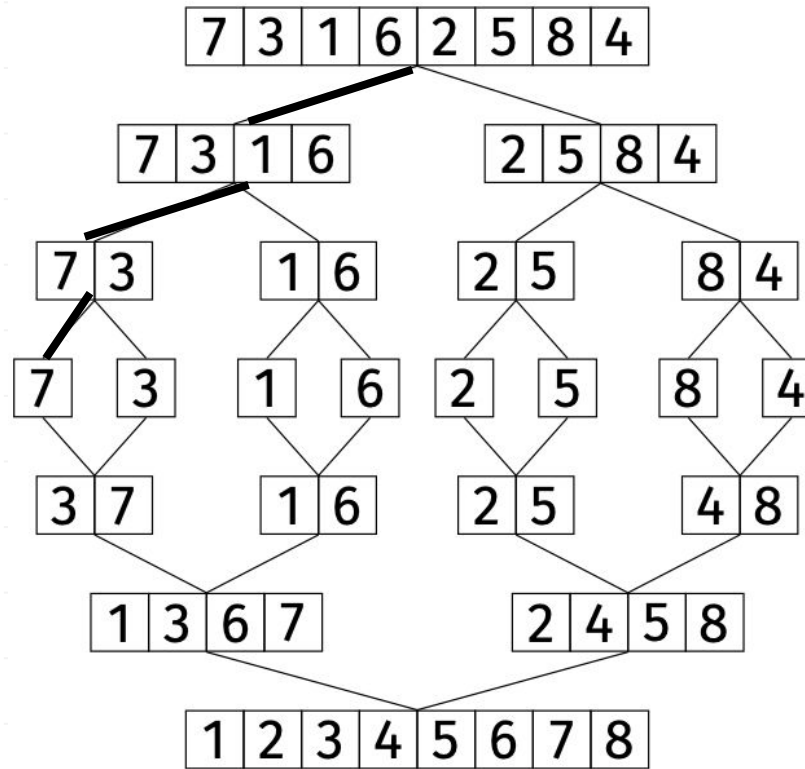


ms([7, 3, 1, 6, 2, 5, 8, 4])

ms([7, 3, 1, 6])

ms([7, 3])

The Idea



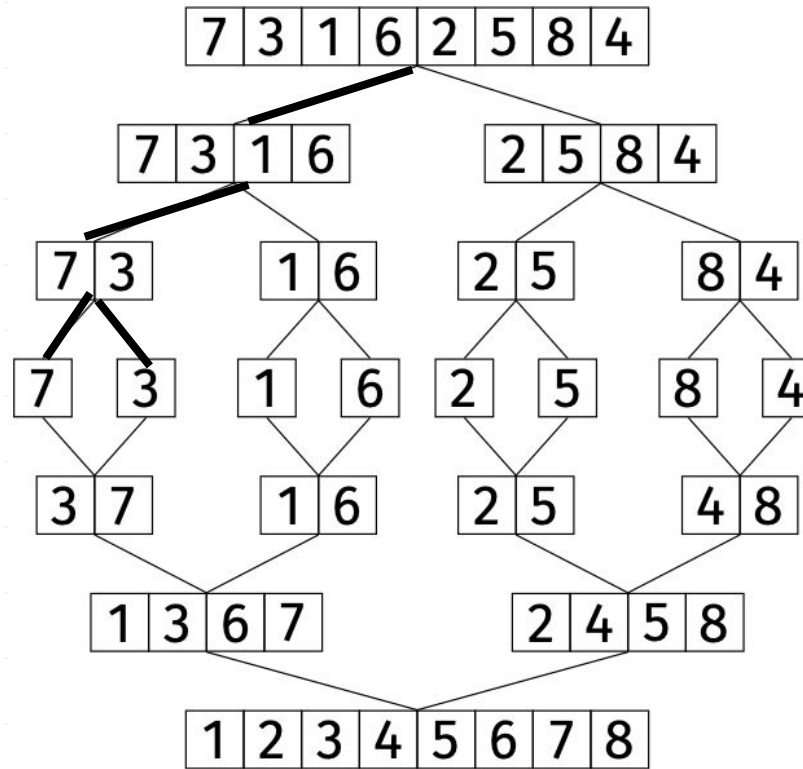
`ms([7, 3, 1, 6, 2, 5, 8, 4])`

`ms([7, 3, 1, 6])`

`ms([7, 3])`

`ms([7])`

The Idea



`ms([7, 3, 1, 6, 2, 5, 8, 4])`

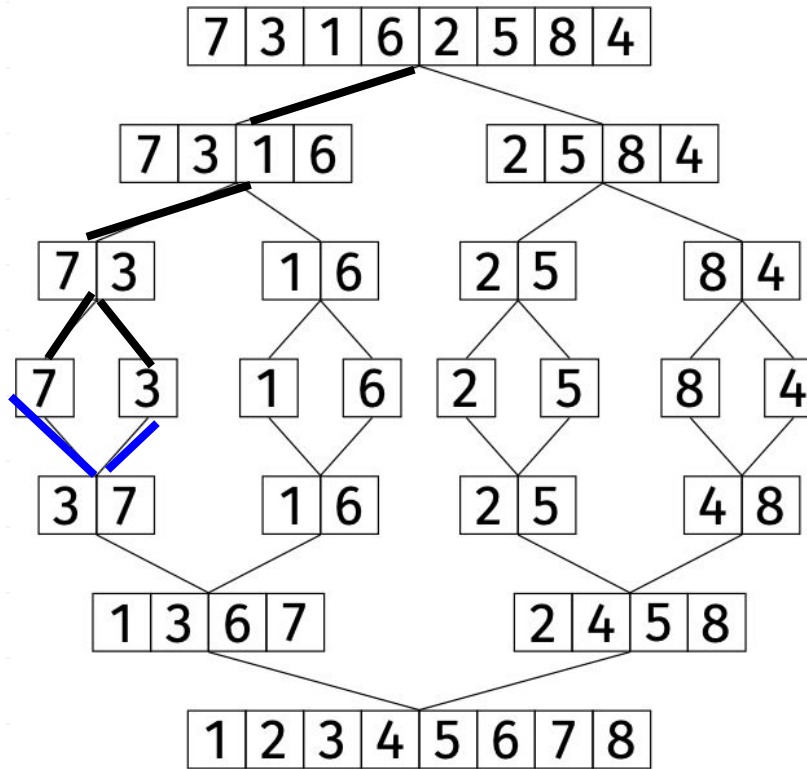
`ms([7, 3, 1, 6])`

`ms([7, 3])`

`ms([7])`

`ms([3])`

The Idea



`ms([7, 3, 1, 6, 2, 5, 8, 4])`

`ms([7, 3, 1, 6])`

`ms([7, 3])`

`ms([7])`

`ms([3])`

`merge([7], [3])`

Mergesort: tip for tracing

```
def mergesort(arr):  
    print(arr)  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        mergesort(left)  
        mergesort(right)  
        merge(left, right, arr)
```

Understanding Mergesort

1. What is the **base** case?
2. Are the recursive problems **smaller**?
3. Assuming the recursive calls work, does the whole algorithm **work**?

1. Base Case: $n = 1$

- Arrays of size one are trivially sorted.
- Returns immediately: **Correct!**

2. Smaller Problems?

- Are `arr[:middle]` and `arr[middle:]` always smaller than `arr`?
- Try it for `len(arr) == 2`.
- **Correct!**

3. Does it Work?

- Assume mergesort works on arrays of size $< n$.
- Does it work on arrays of size n ?

Mergesort

```
def mergesort(arr):  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        mergesort(left)  
        mergesort(right)  
        merge(left, right, arr)
```

Does it work properly?

Merge

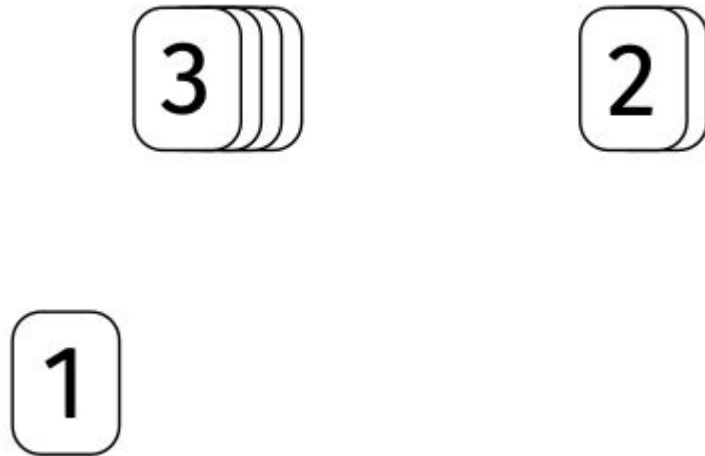
Merging

- We have sorted each half.
- Now we need to **merge** together into a sorted array.
- **Note**: this is an example of a problem that is made easier by sorting.

Merge. Stack of cards in SORTED order



merge



merge

3

6

1

2

merge

5

6

1

2

3

merge

7

6

1

2

3

5

merge

8

1

2

3

5

6

7

merge

1 2 3 5 6 7 8

merge

```
def merge(left, right, out):  
    """Merge sorted arrays, store in out."""  
    left.append(float('inf'))  
    right.append(float('inf'))  
    left_ix = 0  
    right_ix = 0  
  
    for ix in range(len(out)):  
        if left[left_ix] < right[right_ix]:  
            out[ix] = left[left_ix]  
            left_ix += 1  
        else:  
            out[ix] = right[right_ix]  
            right_ix += 1
```

merge

[5, 7, 9, 10]

```
def merge(left, right, out):  
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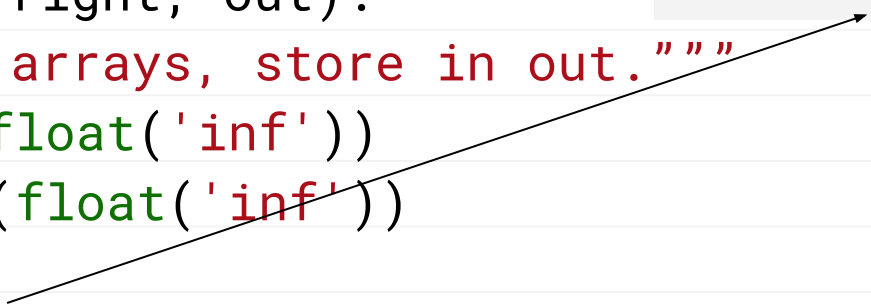
[5, 7, 9, 10, 'inf']



merge

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def merge(left, right, out):  
    """Merge sorted arrays, store in out."""  
    left.append(float('inf'))  
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left = [5, 7, 9, 10, 'inf']



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            out[ix] = right[right_ix]  
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left = [5, 7, 9, 10, 'inf']

right = [2, 4, 6, 'inf']

merge

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def merge(left, right, out):  
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```
    left.append(float('inf'))
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```
    right.append(float('inf'))
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```
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```
    right_ix = 0
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left = [5, 7, 9, 10, 'inf']

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left = [5, 7, 9, 10, 'inf']

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left = [5, 7, 9, 10, 'inf']



right = [2, 4, 6, 'inf']



out = [, , , , ,]

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```

left = [5, 7, 9, 10, 'inf']

right = [2, 4, 6, 'inf']

out = [2, , , , ,]

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def merge(left, right, out):  
    """Merge sorted arrays, store in out."""
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    left.append(float('inf'))
```

```
    right.append(float('inf'))
```

```
    left_ix = 0
```

```
    right_ix = 0
```

left = [5, 7, 9, 10, 'inf']

right = [2, 4, 6, 'inf']

out = [2, 4, , , ,]

```
    for ix in range(len(out)):  
        if left[left_ix] < right[right_ix]:  
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left = [5, 7, 9, 10, 'inf']



right = [2, 4, 6, 'inf']



out = [2, 4, 5, , , ,]

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    for ix in range(len(out)):
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        if left[left_ix] < right[right_ix]:
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            out[ix] = left[left_ix]
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            left_ix += 1
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```
        else:
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            out[ix] = right[right_ix]
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            right_ix += 1
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out = [2, 4, 5, 6, , ,]

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out = [2, 4, 5, 6, 7, ,]

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```

left = [5, 7, 9, 10, 'inf']

right = [2, 4, 6, 'inf']

out = [2, 4, 5, 6, 7, 9,]

```
    for ix in range(len(out)):  
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```
    left_ix = 0
```

```
    right_ix = 0
```

```
    out = [ 2, 4, 5, 6, 7, 9, 10]
```

```
    for ix in range(len(out)):
```

```
        if left[left_ix] < right[right_ix]:
```

```
            out[ix] = left[left_ix]
```

```
            left_ix += 1
```

```
        else:
```

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            out[ix] = right[right_ix]
```

```
            right_ix += 1
```

```
left = [5, 7, 9, 10, 'inf']
```



```
right = [2, 4, 6, 'inf']
```



Loop Invariant

- Assume `left` and `right` are sorted.
- **Loop invariant:** After α th iteration, first α elements of `out` are the smallest α elements of those in `left` and `right`, in sorted order.
- That is, after α th iteration

`out[: $\alpha$ ] == sorted(left + right)[: $\alpha$ ]`

Key of mergesort

- merge is where the **actual sorting** happens.
- **Example:** merge([3], [1], ...) results in [1,3]

Merge: time complexity

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def merge(left, right, out):  
    """Merge sorted arrays, store in out."""  
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    right.append(float('inf'))  
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**Time complexity for
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```

Time complexity for merge?

A: Constant

B: Log

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Time Complexity of Mergesort

Mergesort: time complexity

```
def mergesort(arr):  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        mergesort(left)  
        mergesort(right)  
        merge(left, right, arr)
```

$T(n) = \text{non_recursive} + \text{recursive}$

Mergesort: time complexity

```
def mergesort(arr):
```

```
    if len(arr) > 1:
```

```
        middle = math.floor(len(arr) / 2)
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```
        left = arr[:middle]
```

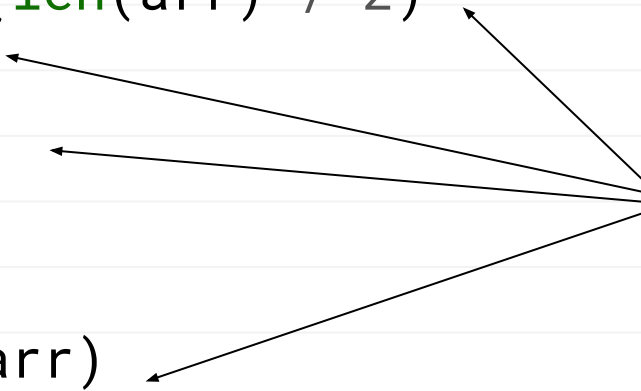
```
        right = arr[middle:]
```

```
        mergesort(left)
```

```
        mergesort(right)
```

```
        merge(left, right, arr)
```

$T(n) = \text{non_recursive} + \text{recursive}$



Mergesort: time complexity

```
def mergesort(arr):
```

```
    if len(arr) > 1:
```

```
        middle = math.floor(len(arr) / 2)
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```
        left = arr[:middle]
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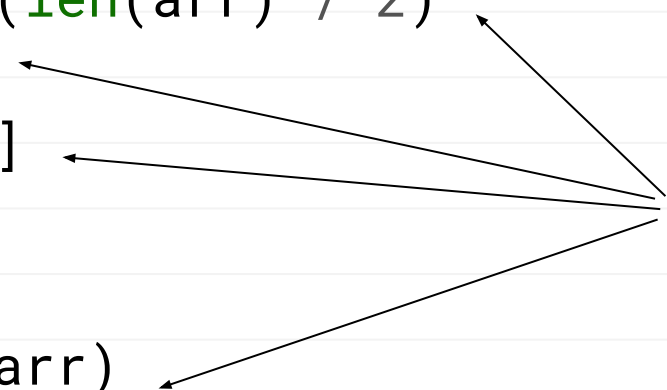
```
        mergesort(left)
```

```
        mergesort(right)
```

```
        merge(left, right, arr)
```

$T(n) = \text{non_recursive} + \text{recursive}$

$\Theta(n)$



Mergesort: time complexity

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    if len(arr) > 1:  
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Mergesort: time complexity

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def mergesort(arr):  
    if len(arr) > 1:  
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        left = arr[:middle]  
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        mergesort(left) ←  
        mergesort(right) ←  
        merge(left, right, arr)
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$T(n) = \Theta(n) + \text{recursive}$

Mergesort: time complexity

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def mergesort(arr):
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```
    if len(arr) > 1:
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```
        middle = math.floor(len(arr) / 2)
```

```
        left = arr[:middle]
```

```
        right = arr[middle:]
```

```
        mergesort(left) ← T(n/2)
```

```
        mergesort(right) ← T(n/2)
```

```
        merge(left, right, arr)
```

$T(n) = \Theta(n) + \text{recursive}$

Mergesort: time complexity

```
def mergesort(arr):  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        mergesort(left) ← T(n/2)  
        mergesort(right) ← T(n/2)  
        merge(left, right, arr)
```

$$T(n) = \Theta(n) + 2 T(n/2)$$

Solving the Recurrence

$$T(n) = 2 T(n/2) + \Theta(n)$$

Solving the Recurrence. Step 2

$$T(n) = 2 T(n/2) + n \quad \# \mathbf{k=1}$$

$$T\left(\frac{n}{2}\right) = 2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2}$$

Solving the Recurrence. Step 2

$$T(n) = 2 T(n/2) + n \quad \# \mathbf{k=1}$$

$$T\left(\frac{n}{2}\right) = 2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2}$$

$$T(n) = 2 \cdot \left[2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2} \right] + n$$

Solving the Recurrence. Step 2

$$T(n) = 2 T(n/2) + n \quad \# \mathbf{k=1}$$

$$T\left(\frac{n}{2}\right) = 2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2}$$

$$T(n) = 2 \cdot \left[2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2} \right] + n \quad \# \text{ simplify}$$

Solving the Recurrence. Step 2

$$T(n) = 2 T(n/2) + n \quad \# \mathbf{k=1}$$

$$T\left(\frac{n}{2}\right) = 2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2}$$

$$T(n) = 2 \cdot \left[2 \cdot T\left(\frac{n}{4}\right) + \frac{n}{2} \right] + n \quad \# \text{ simplify}$$

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$$T(n) = 4 \cdot \left[2 \cdot T\left(\frac{n}{8}\right) + \frac{n}{4} \right] + 2n \quad \# \text{ simplify}$$

$$T(n) = 8 \cdot T\left(\frac{n}{8}\right) + 3n \quad \# \mathbf{k=3}$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

Step 3. Number of steps to Base case.

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$$\frac{n}{2^k} = 1 \quad n = 2^k \quad \Rightarrow \quad k = \log_2 n$$

Step 4. Substitute

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

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Step 4. Substitute

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$\frac{n}{2^k} = 1 \quad n = 2^k \Rightarrow k = \log_2 n$$

$$T(n) = 2^{\log_2 n} \cdot T\left(\frac{n}{2^{\log_2 n}}\right) + n \log_2 n$$

$$\Theta(n \log n)$$

$$T(n) = n \cdot T(1) + n \log_2 n$$

Optimality

- **Theorem:** Any (comparison) sorting algorithm's worst-case time complexity must be $\Omega(n \log n)$.
- Mergesort is **optimal**!

Be Careful!

- It is possible for a sorting algorithm to have a **best case** time complexity **smaller than $n \log n$** .
 - Insertion sort, for example.
- Mergesort has best case time complexity of **$\Theta(n \log n)$** .
- Mergesort is **sub-optimal** in this sense!

Be Careful!

- The $\Theta(n \log n)$ lower-bound is for **comparison sorting**.
- It is possible to sort in worst-case $\Theta(n)$ time without comparing.
 - Bucket sort, radix sort, etc.

What if?

- **Divide**: split the array into halves
- **Conquer**: sort each half using **selection sort**
- **Combine**: merge sorted halves together

mergeselectionsort

```
def mergeselectionsort(arr):  
    """Sort array in-place."""  
    if len(arr) > 1:  
        middle = math.floor(len(arr) / 2)  
        left = arr[:middle]  
        right = arr[middle:]  
        selection_sort(left)  
        selection_sort(right)  
        merge(left, right, arr)
```

Think!

What is the time complexity of this algorithm?

Using Sorted Structure

Sorted structure is useful

- Some problems become much easier if input is sorted.
- For example, median, minimum, maximum.
- Sorting is useful as a preprocessing step.

Recall: The Movie Problem

- You're on a flight that will last D minutes.
- You want to pick two movies to watch.
- You want the total time of the two movies to be as close as possible to D .

The Movie Problem

- Brute force algorithm: $\Theta(n^2)$
- We can do better, if movie times are **sorted**.

Example

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

	60	80	90	120	130
60					
80					
90					
120					
130					

Best pair:

Example: with brute force only half

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

	60	80	90	120	130
60					
80	x				
90	x	x			
120	x	x	x		
130	x	x	x	x	

Best pair:

Example: with brute force only half

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

	60	80	90	120	130
60	x				
80	x	x			
90	x	x	x		
120	x	x	x	x	
130	x	x	x	x	x

Best pair:

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x				+35
80	x	x			
90	x	x	x		
120	x	x	x	x	
130	x	x	x	x	x

Best pair: 60, 130
+35 above the target

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x				+35
80	x	x			
90	x	x	x		
120	x	x	x	x	
130	x	x	x	x	x

Best pair: 60, 130
+35 above the target

A: 80, 130

B: 90, 130

C: 120, 130

D: 60, 120

E: I have no idea

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x				+35
80	x	x			x
90	x	x	x		x
120	x	x	x	x	x
130	x	x	x	x	x

Best pair: 60, 130
+35 above the target

A: 80, 130

B: 90, 130

C: 120, 130

D: **60, 120**

E: I have no idea

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x			+25	+35
80	x	x			x
90	x	x	x		x
120	x	x	x	x	x
130	x	x	x	x	x

Best pair: 60, 120
+25 above the target

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x			+25	+35
80	x	x		x	x
90	x	x	x	x	x
120	x	x	x	x	x
130	x	x	x	x	x

Best pair: 60, 120
+25 above the target

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x		-5	+25	+35
80	x	x		x	x
90	x	x	x	x	x
120	x	x	x	x	x
130	x	x	x	x	x

Best pair: 60, 90
-5 below the target

Example:

- Flight duration $D = 155$
- Movie times: 60, 80, 90, 120, 130

First step: take
shortest + longest.

	60	80	90	120	130
60	x	x	-5	+25	+35
80	x	x		x	x
90	x	x	x	x	x
120	x	x	x	x	x
130	x	x	x	x	x

Best pair: 60, 90
-5 below the target

The Algorithm

- Keep *index* of shortest and longest remaining movies.
- On every iteration, pair the shortest and longest.
- If this pair is too long, remove longest movie; otherwise remove shortest.
 - If times are **sorted**, finding new longest/shortest movie takes $\Theta(1)$ time!

60, 80, 90, 120, 130

```
def optimize_entertainment(times, target):  
    """assume times is sorted."""  
    shortest = 0  
    longest = len(times) - 1  
  
    best_pair = (shortest, longest)  
    best_objective = None  
  
    for i in range(len(times) - 1):  
        total_time = times[shortest] + times[longest]  
  
        if abs(total_time - target) < best_objective:  
            best_objective = abs(total_time - target)  
            best_pair = (shortest, longest)  
  
        if total_time == target:  
            return (shortest, longest)  
        elif total_time < target:  
            shortest += 1  
        else: # total_time > target  
            longest -= 1  
  
    return best_pair
```

Main Idea

Sorted structure allows you to **rule out** possibilities without explicitly checking them. But, it requires you to **spend the time sorting first.**

Tip: when designing an algorithm, think about sorting the input first.



Thank you!

Do you have any questions?

CampusWire!